Development of Machine Learning Algorithms for Automatic Geotag Recognition in Textual Data

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Abstract

Geographic information extraction from unstructured textual data has become increasingly critical in domains ranging from smart city management to emergency response systems. This thesis presents a comparative study of machine learning approaches for automatic geotag recognition — a form of Named Entity Recognition (NER) — applied to a real-world multilingual dataset of public transport complaint records in Kazakh and Russian languages, provided by City Transportation Systems (CTS) of Astana, Kazakhstan.

Four distinct approaches are developed and evaluated: (1) fine-tuned XLM-RoBERTa, (2) fine-tuned ruBERT, (3) a few-shot Large Language Model pipeline based on Qwen2.5 (7B) deployed locally via Ollama, and (4) a fine-tuned Multilingual-E5-base (Me5-base) encoder. The dataset comprises 2,750 manually annotated complaint records with six domain-specific entity types: `ROUTE_NUM`, `BUS_ID`, `PLATE_NUM`, `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION`, annotated using the BIO tagging scheme.

Experimental results demonstrate that the Qwen2.5 LLM achieves the highest overall precision (0.788) and macro F1-score (0.635), and is the only model capable of detecting the complex `STREET_INTERSECTION` entity. Me5-base achieves the best recall (0.653). XLM-RoBERTa provides the strongest single-entity performance on `ROUTE_NUM` (F1 = 0.920), while ruBERT (F1 = 0.223) is shown to be inadequate for code-switching multilingual data. These findings provide practical guidance for deploying geotag recognition systems in low-resource, multilingual operational environments.

Keywords: Named Entity Recognition, Geotag Extraction, XLM-RoBERTa, BERT, Large Language Models, Multilingual NLP, Kazakh, Russian, Transport Complaints, BIO Tagging, Sequence Labeling

Author's declaration

I declare that the work in this diploma work was carried out in accordance with the requirements of the University's Regulations and that it has not been submitted for any other academic award. Except where indicated by specific references in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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Introduction

The rapid digitisation of urban services has produced an ever-growing volume of unstructured text. Public transport operators, emergency services, and municipal authorities now receive thousands of complaint messages, incident reports, and feedback submissions every day. Buried within these free-form texts are geographic references — street names, transit stop names, route numbers, and intersection descriptions — that, if extracted automatically, could power real-time situational awareness dashboards, smart route-optimisation engines, and automated incident-routing systems.

This thesis addresses the problem of *automatic geotag recognition*: the automatic identification and classification of geographically meaningful textual spans within domain-specific, multilingual operational text. The work was motivated by a concrete industrial need posed by City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan, which provided a corpus of approximately 3 000 complaint records as the primary data source.

The research makes the following original contributions:

1. A custom-annotated, domain-specific NER dataset of 2 750 multilingual (Kazakh/Russian) transit complaint records with six entity types.

2. Systematic evaluation of four machine learning approaches for geotag NER: fine-tuned XLM-RoBERTa, fine-tuned ruBERT, few-shot Qwen2.5 LLM inference, and fine-tuned Multilingual-E5-base.

3. Entity-level performance analysis with confusion matrices, identifying failure modes specific to code-switching transit data.

4. Practical deployment recommendations for real-time geotag recognition in resource-constrained multilingual environments.

The remainder of the thesis is organised as follows. Chapter 1 reviews the relevant literature and formulates the research problem. Chapter 2 describes the dataset, annotation methodology, and experimental setup. Chapter 3 presents results, discussion, and conclusions.

Chapter 1

Literature Review and Problem Statement

1.1 Background: Named Entity Recognition

Named Entity Recognition (NER) is a fundamental natural language processing task that involves identifying and classifying textual spans into predefined semantic categories such as persons, organisations, locations, and domain-specific types [?]. NER serves as a foundational component for downstream tasks including relation extraction, knowledge graph construction, question answering, and — most relevant to this work — geospatial information retrieval.

Early NER systems relied on handcrafted rules and statistical models such as Hidden Markov Models (HMM) and Maximum Entropy classifiers. The introduction of Conditional Random Fields (CRF) [?] substantially improved sequence labelling performance by modelling label dependencies globally. Deep learning approaches, particularly BiLSTM-CRF architectures [?], automated feature extraction and set new benchmarks on standard corpora such as CoNLL-2003.

The transformer revolution, initiated by Vaswani et al. [?] and popularised for NLP through BERT [?], established pre-trained contextual embeddings as the dominant paradigm. Fine-tuning BERT-style encoders on labelled NER data quickly became the default approach, with models such as RoBERTa-large achieving F1 = 0.93 on CoNLL-2003 [?].

1.2 Multilingual and Cross-Lingual NER

For multilingual settings, cross-lingual pre-trained models are essential. XLM-RoBERTa [?], trained on 2.5 TB of CommonCrawl data across 100 languages, enables zero-shot and fine-tuned NER transfer across language boundaries. Iqbal et al. [?] demonstrate that multilingual BERT achieves macro F1 = 0.98 on an augmented multilingual NER dataset, outperforming monolingual RoBERTa (F1 = 0.884). Kassab et al. [?] show that DeepPavlov RuBERT remains the strongest single Russian-language NER model, while semi-automated ensemble pipelines improve precision by 28% over the single-model baseline.

1.3 Geoparsing and Geographic NER

Geoparsing — the automatic recognition and coordinate resolution of geographic entities in text — extends standard NER with spatial disambiguation. Hu et al. [?] evaluate 27 geoparsing tools across 26 public datasets (tweets, news, scientific articles), finding that BERT-based models achieve the highest F1 on social media text (TopoBERT F1 = 0.854), while classical tools degrade sharply on noisy input. Moncla et al. [?] address toponym ambiguity resolution using CamemBERT and XLM-RoBERTa, reporting F1 $\approx$ 0.92 on diverse geographic corpora. Tao et al. [?] show that enhanced BERT + BiLSTM + CRF achieves F1 = 0.93 on geographically annotated social media data versus 0.78 for standard BiLSTM-CRF.

1.4 Large Language Models for Information Extraction

Recent work has explored whether large language models (LLMs) can replace fine-tuned encoders for NER. Keraghel et al. [?] find that GPT-4 in zero-shot mode achieves F1 = 0.83 on CoNLL-2003, compared to 0.93 for fine-tuned RoBERTa-large, but LLMs excel in few-shot transfer settings. Yang et al. [?] present the Qwen2.5 series, demonstrating that instruction tuning significantly improves structured output generation, including JSON extraction. Mai et al. [?] show that LLMs fused with geographic knowledge bases substantially improve geoparsing precision and fairness for underrepresented regions. Lutsai and Lampert [?] achieve 96% accuracy with a median spatial error of

3.68 km on 150 million geo-tagged tweets using a multitask PMOP regression model with multilingual BERT.

A broader literature survey covering all 20 referenced works is summarised in Table 1.1.

Table 1.1: Summary of key related works.

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
1	Yang et al. (2023) [?]	BERT, GPT-2/3/4, T5, LLaMA, BLOOM	Survey (no dataset)	Covers evolution from BERT to ChatGPT across pretraining objectives and architectures
2	Keraghel et al. (2024) [?]	BiLSTM-CRF, BERT, RoBERTa, GPT-3/4, LLaMA	RoBERTa F1 = 0.93; GPT-4 zero-shot = 0.83	Fine-tuned encoders dominate supervised NER; LLMs excel in few-shot settings
3	Hu et al. (2023) [?]	27 tools incl. TopoBERT, spaCy, Stanford NER	TopoBERT F1 = 0.854	BERT-based models best on social media; classical tools degrade on noisy text
4	Iqbal et al. (2024) [?]	mBERT, RoBERTa-small/large, BERT-base/large	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa; larger size does not always help
5	Kassab et al. (2025) [?]	DeepPavlov RuBERT, Natasha, Stanza, spaCy	+28% precision over single RuBERT	Semi-automated multi-model voting reduces annotation effort drastically
6	Petasis & Karkaletsis (2024) [?]	E5-large + XGBoost, KNN, LogReg (soft voting)	Accuracy = 0.92; F1 = 0.91	E5 embeddings + ML ensemble beats fine-tuned transformers on low-resource language
7	Wang et al. (2023) [?]	mE5-small/base/large/large-instruct (XLM-R)	large-instruct = 64.4	Two-stage training yields strong multilingual embeddings; degrades on low-resource languages
8	Yang et al. (2024) [?]	Qwen2.5 (0.5B–72B, Base, Instruct, MoE)	MMLU, HumanEval, MATH	Instruction tuning improves structured output; open-weight models are cost-effective
9	Mai et al. (2024) [?]	Mistral, Baichuan2, Llama 2, Falcon	Improved precision & fairness	LLM fused with geo-knowledge improves geoparsing for underrepresented regions
10	Tarannum et al. (2026) [?]	mBERT, RoBERTa-small, BERT-large/base	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa on augmented address data
11	Wei et al. (2024) [?]	ALBERT-Conv1D-5 BiLSTM-CRF	P = 0.981; R = 0.967; F1 = 0.974	Fusing global (ALBERT) and local (Conv1D) features improves historical toponym NER
12	Moncla et al.	CamemBERT,	CamemBERT F1	Neural classifiers on

1.5 Research Gap and Problem Statement

Despite substantial progress in multilingual NER and geoparsing, several gaps remain unaddressed in existing literature. First, no prior work targets the specific domain of public transport complaint texts in the Kazakh-Russian code-switching context, a linguistic setting distinct from the English-dominant benchmarks used in most studies. Second, the majority of geospatial NER research focuses on named locations and administrative regions, while operational entities such as route numbers, vehicle plate numbers, and street intersections — critical for transit management — remain understudied. Third, there is no systematic comparison of fine-tuned encoder models against few-shot LLM inference on a domain-specific, low-resource multilingual corpus.

1.5.1 Formal Problem Definition

We formulate geotag recognition as a token-level sequence labelling task under the BIO (Beginning-Inside-Outside) tagging scheme. Given an input token sequence $\mathbf{T} = \{t_1, t_2, \ldots, t_n\}$, the objective is to assign each token t_i a label $l_i \in \mathcal{L}$, where:

$$\mathcal{L} = \{B\text{-}e,\ I\text{-}e \mid e \in \mathcal{E}\} \cup \{O\}$$

$\mathcal{E}$ denotes the set of entity types (defined in Section 1.5.2), and O denotes the outside (non-entity) class. The BIO scheme encodes entity spans such that the first token of an entity receives the B- prefix, subsequent tokens within the same entity receive the I- prefix, and all non-entity tokens receive O.

1.5.2 Entity Types

Based on domain requirements and corpus analysis, six entity types were defined:

`ROUTE_NUM`	Public transit route number (e.g., «Маршрут 47», «Автобус № 64»).
`BUS_ID`	Vehicle identifier or fleet number assigned to a specific bus.
`PLATE_NUM`	Vehicle licence plate number.
`STOP_NAME`	Named transit or bus stop (e.g., «аялдама Қобыланды»).
`STREET_NAME`	Named street, avenue, or road.

`STREET_INTERSECTION` Geographic intersection of two or more streets; the primary geospatial geotag entity.

1.5.3 Research Aim and Objectives

Research Aim: To develop and evaluate multiple machine learning approaches for automatic geotag recognition in multilingual textual data and to compare their effectiveness in terms of accuracy and robustness.

Objectives:

1. Analysis of current methods for automatic geotag recognition (literature review).
2. Collection and preprocessing of a domain-specific multilingual dataset with geographic annotation.
3. Design and implementation of four machine learning–based geotag recognition approaches.
4. Evaluation and comparison of model performance using standard NER metrics (Precision, Recall, F1).
5. Analysis of results and identification of the most effective approach for deployment.

Chapter 2

Dataset, Methodology, and Experimental Results

2.1 Dataset and Annotation

2.1.1 Data Source

The dataset was provided directly by City Transportation Systems (CTS), the public transport operator of Astana, Kazakhstan. The raw corpus consists of approximately 3 000 complaint records collected through operational channels including a mobile application, a web portal, and call-centre transcriptions. Each record contains a free-text description field (*описание*) in which passengers describe transportation issues: missed stops, route deviations, driver misconduct, and payment failures.

2.1.2 Annotation Procedure

Manual annotation was performed using *Label Studio*, an open-source data-labelling platform. Three annotators with native-level proficiency in both Kazakh and Russian independently labelled each complaint record with the six entity types defined in Section 1.5.2. Annotation guidelines were developed iteratively, with inter-annotator disagreements resolved through adjudication to produce a gold-standard label set.

The final annotated dataset comprises 2 750 complaint records exported in JSONL format with BIO token-level labels, split into:

- **Training set:** 1 925 samples (70 %), used for model fine-tuning.
- **Test set:** 825 samples (30 %), used exclusively for evaluation.

Stratified sampling was applied to maintain entity type proportions across splits. Table 2.1 presents the dataset statistics.

Table 2.1: Dataset statistics by entity type.

Entity Type	**Train Tokens**	**Test Tokens**	**Total**	**Avg./Record**
`ROUTE_NUM`	2,641	1,131	3,772	1.37
`BUS_ID`	798	342	1,140	0.41
`PLATE_NUM`	623	267	890	0.32
`STOP_NAME`	3,901	1,672	5,573	2.03
`STREET_NAME`	287	123	410	0.15
`STREET_INTERSECTION`	52	22	74	0.03
Total	8,302	3,557	11,859	–

2.1.3 Data Characteristics and Challenges

The corpus presents several challenges relevant to model selection:

1. **Code-switching.** Approximately 67 % of records contain intra-sentence Kazakh-Russian mixing. Complainants naturally alternate between languages within a single message.

2. **Informal register.** Punctuation is sparse, abbreviations are domain-specific, and spelling inconsistencies are common.

3. **Class imbalance.** `STOP_NAME` accounts for 2.03 entities per record on average, while `STREET_INTERSECTION` appears only 0.03 times per record — a ratio of approximately 75:1.

4. **Structural complexity of intersections.** `STREET_INTERSECTION` entities typically span 5–10 tokens including conjunction words (*угол*, *пересечение*), two distinct street names, and optional qualifiers.

2.2 Methodology

We evaluate four machine learning approaches. The BIO label set comprises 13 entity labels (B- and I- for each of the 6 entity types) plus the O class, yielding $K = 14$

output classes. All fine-tuning experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM).

2.2.1 XLM-RoBERTa

XLM-RoBERTa (`xlm-roberta-base`) [?] is a multilingual transformer encoder pre-trained on 2.5 TB of CommonCrawl data across 100 languages using a masked language modelling objective. Its cross-lingual transfer capabilities and coverage of both Kazakh and Russian make it well-suited for the code-switching characteristics of the CTS corpus.

Architecture. The base encoder (12 transformer layers, 768 hidden dimensions, 12 attention heads, ≈125M parameters) is extended with a token classification head, a linear layer $W \in \mathbb{R}^{768\times K}$ projecting each contextual embedding to class logits. Token probabilities are computed via the softmax function:

$$p(y_i = c \mid \mathbf{x}_i) = \frac{\exp(z_{i,c})}{\sum_{k=1}^{K} \exp(z_{i,k})} \tag{2.1}$$

where $\mathbf{x}_i$ is the contextual embedding of token i, $z_{i,c}$ is the raw logit for class c, and $K = 14$.

Loss function. Training uses weighted cross-entropy to address class imbalance:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i} \log p(y_i \mid \mathbf{x}_i), \qquad w_c = \frac{N_{\text{total}}}{K \cdot N_c} \tag{2.2}$$

where N is the number of valid (non-padding) tokens, N_{total} is the total number of labelled tokens, and N_c is the frequency of class c. The `STREET_NAME` class receives an additional manual upweight factor of 1.6.

Learning rate schedule. A linear decay is applied:

$$\text{lr}(t) = \text{lr}_{\max} \cdot \left(1 - \frac{t}{T}\right) \tag{2.3}$$

where $\text{lr}_{\max} = 2 \times 10^{-5}$, t is the current training step, and T is the total number of steps.

Training configuration: 10 epochs, batch size 8, AdamW optimiser (weight decay 0.01), SentencePiece BPE tokeniser (vocabulary size 250 000).

2.2.2 ruBERT

ruBERT (`DeepPavlov/rubert-base-cased`) [?] is a BERT-base model pre-trained on Russian Wikipedia and news corpora (≈180M parameters). It serves as a monolingual Russian-language baseline.

Token label alignment. Sub-word tokenisation is handled via offset mapping:

$$L'_i = \begin{cases} L_i & \text{if offset}_i \neq (0,0) \\ -100 & \text{otherwise} \end{cases} \tag{2.4}$$

where L_i is the original BIO label and -100 masks special/padding tokens from the loss.

Prediction. Token label assignment follows:

$$\hat{y}_i = \arg\max_j \text{softmax}(W\mathbf{x}_i + \mathbf{b})_j \tag{2.5}$$

Training configuration: 3 epochs, batch size 8, learning rate 2×10^{-5}, max sequence length 128, AdamW optimiser, unweighted cross-entropy loss.

2.2.3 LLM Few-Shot Inference (Qwen2.5:7B)

Qwen2.5 [?] is Alibaba's open-source 7B-parameter instruction-tuned LLM supporting 29+ languages including Russian and Kazakh. The model was deployed locally via the *Ollama* serving framework, enabling inference without external API calls.

Prompt construction. The full inference prompt is constructed as:

$$\text{prompt}_i = \text{SYS} \oplus E_{1..12} \oplus x_i \tag{2.6}$$

where SYS is a system instruction defining the 6 entity types and extraction rules for Kazakh/Russian transit complaints, $E_{1..12}$ are 12 balanced in-context examples (2 per entity type) from the training set, and x_i is the raw complaint text.

Post-filtering. To remove hallucinated spans, a post-filter is applied:

$$\hat{\mathcal{E}}_i^{(c)} = \left\{ v \in \hat{\mathcal{E}}_i^{(c)} \mid |v| \geq 2,\ v \notin \mathcal{W}_{\text{stop}},\ \phi(x_i, c) \right\} \tag{2.7}$$

where $|v|$ is the character length, $\mathcal{W}_{\text{stop}}$ is a set of noise words, and $\phi(x_i, c)$ is a keyword check required for `STREET_INTERSECTION` entities.

Inference configuration: max tokens 300, timeout 90 s / 3 retries, seed 42, 12 few-shot examples. Evaluated on the first 300 test samples due to computational constraints.

2.2.4 Multilingual-E5-base (Me5-base)

Multilingual-E5-base (`intfloat/multilingual-e5-base`) [?] is an XLM-RoBERTa-based sentence embedding model (≈278M parameters) trained using contrastive learning on ≈1 billion multilingual text pairs. A token classification head is attached for NER:

$$Z = XW + b \tag{2.8}$$

where $X \in \mathbb{R}^{N\times 768}$ is the matrix of contextual token embeddings and $Z \in \mathbb{R}^{N\times 13}$ contains output logits.

Class weights are computed as:

$$w_c = \frac{\sum_{j\in\text{non-O}} \text{count}_j}{|\text{non-O}| \cdot \text{count}_c} \tag{2.9}$$

clipped to [0.05, 8.0] for numerical stability.

Training configuration: 15 epochs, batch size 8 (train) / 16 (eval), gradient accumulation 2 steps, learning rate 2×10^{-5}, AdamW (weight decay 0.01, warmup ratio 0.1), seed 42.

2.3 Evaluation Metrics

All models are evaluated using standard span-level NER metrics computed by the `seqeval` library:

$$\text{Precision} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}|}, \quad \text{Recall} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2.10}$$

Macro-averaged F1 weights each entity type equally, regardless of frequency, making it sensitive to performance on rare classes.

2.4 Experimental Results

2.4.1 XLM-RoBERTa Results

XLM-RoBERTa demonstrates the strongest performance on high-frequency, structurally regular entities. `ROUTE_NUM` achieves near-state-of-the-art F1 = 0.920, attributed to the consistent numeric format of route identifiers and their high training frequency. `BUS_ID` performance is moderate (F1 = 0.735), while `PLATE_NUM` and `STOP_NAME` achieve comparable scores (≈0.633–0.650). `STREET_NAME` achieves only F1 = 0.294 due to class imbalance and morphological variation. `STREET_INTERSECTION` receives zero detections (F1 = 0.000), reflecting extreme scarcity and structural complexity.

Table 2.2: XLM-RoBERTa entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.884	0.960	0.920
BUS_ID	0.889	0.626	0.735
PLATE_NUM	0.527	0.847	0.650
STOP_NAME	0.514	0.826	0.633
STREET_NAME	0.392	0.236	0.294
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.534	0.582	**0.539**

2.4.2 ruBERT Results

ruBERT exhibits severely degraded performance (macro F1 = 0.223), consistent with the code-switching nature of the corpus. The model fails entirely on BUS_ID, STREET_NAME, and STREET_INTERSECTION (F1 = 0.000 for all three), and achieves only F1 = 0.667 for ROUTE_NUM. Analysis of the confusion matrix reveals systematic label collapse toward STOP_NAME, the most frequent class. This behaviour is a known failure mode of monolingual models applied to out-of-vocabulary code-switching text.

Table 2.3: ruBERT entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.721	0.614	0.667
BUS_ID	0.000	0.000	0.000
PLATE_NUM	0.225	0.343	0.284
STOP_NAME	0.403	0.371	0.387
STREET_NAME	0.000	0.000	0.000
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.225	0.221	**0.223**

2.4.3 LLM Qwen2.5:7B Results

The Qwen2.5 few-shot pipeline achieves the highest overall precision (0.788) and macro F1-score (0.635). It is the only model to detect STREET_INTERSECTION enti-

ties (F1 = 0.667, Precision = 1.000, Recall = 0.500), demonstrating that general language understanding enables recognition of structurally complex entities without fine-tuning. `ROUTE_NUM` (F1 = 0.913) and `STOP_NAME` (F1 = 0.716) outperform all fine-tuned encoder models. Among the four LLMs evaluated (Qwen2.5:7B, Mistral:7B, LLaMA3.1:8B, Gemma2:2B), Qwen2.5 dominates on all three metrics, as shown in Table 2.5.

Table 2.4: Qwen2.5:7B entity-level performance (test set, $n = 300$).

Entity	**Precision**	**Recall**	**F1-Score**
`ROUTE_NUM`	0.890	0.938	0.913
`BUS_ID`	0.953	0.394	0.558
`PLATE_NUM`	0.525	0.913	0.667
`STOP_NAME`	0.690	0.745	0.716
`STREET_NAME`	0.667	0.185	0.289
`STREET_INTERSECTION`	1.000	0.500	0.667
Macro average	0.788	0.613	**0.635**

Table 2.5: LLM comparison: macro-averaged metrics on 300 test samples.

Model	**Avg Precision**	**Avg Recall**	**Avg F1**
Qwen2.5:7B	**0.788**	0.612	**0.635**
Mistral:7B	0.601	0.497	0.513
LLaMA3.1:8B	0.611	0.441	0.483
Gemma2:2B	0.335	**0.569**	0.399

2.4.4 Me5-base Results

Me5-base achieves the highest recall (0.653) and competitive F1-score (0.608). `ROUTE_NUM` performance is near-identical to XLM-RoBERTa (F1 = 0.922). `BUS_ID` (F1 = 0.692) and `STOP_NAME` (F1 = 0.610) show improved recall. The model fails on `STREET_INTERSECTION` (F1 = 0.000), consistent with the encoder baselines.

Table 2.6: Me5-base entity-level performance (test set, $n = 300$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.889	0.957	0.922
BUS_ID	0.879	0.570	0.692
PLATE_NUM	0.502	0.869	0.636
STOP_NAME	0.498	0.787	0.610
STREET_NAME	0.304	0.135	0.187
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.612	0.653	**0.608**

2.4.5 Overall Model Comparison

Table 2.7 summarises the macro-averaged performance across all four approaches. Figure 2.1 illustrates per-entity F1 scores.

Table 2.7: Overall model comparison (macro-averaged metrics).

Model	**Precision**	**Recall**	**F1**	**STREET_INTER. F1**
XLM-RoBERTa	0.534	0.582	0.539	0.000
ruBERT	0.225	0.221	0.223	0.000
Qwen2.5:7B (LLM)	**0.788**	0.613	**0.635**	**0.667**
Me5-base	0.612	**0.653**	0.608	0.000

Per-entity F1 score comparison across all four models (XLM-RoBERTa, ruBERT, Qwen2.5:7B, Me5-base).

To be replaced with pgfplots bar chart.

Figure 2.1: Per-entity F1 scores for all evaluated models on the test set.

2.5 Discussion

2.5.1 Impact of Multilingual Pre-training

The most striking finding of this study is the catastrophic failure of ruBERT (macro F1 = 0.223) on the code-switching corpus, despite the Russian-dominant character of the text. The 67 % rate of intra-sentence Kazakh-Russian mixing is sufficient to severely degrade a monolingual model. This strongly supports the use of massively multilingual pre-trained models (XLM-RoBERTa, mE5) for any NER application in post-Soviet multilingual settings.

2.5.2 Fine-Tuning vs. Few-Shot LLM Inference

The superior overall performance of Qwen2.5 despite receiving no gradient updates raises important practical considerations. The LLM advantage stems from: (1) its 7B parameter scale enabling richer compositional reasoning for complex entities; (2) instruction tuning on diverse multilingual data; and (3) entity-level JSON output, which bypasses BIO boundary errors that systematically penalise fine-tuned models.

However, fine-tuned encoders retain advantages for structurally regular entities (`ROUTE_NUM` F1 $\approx$ 0.92) and offer orders-of-magnitude better throughput: processing the full 825-sample test set takes $\approx$12–18 seconds on GPU for encoders, versus 4+ hours

for 300 Qwen2.5 samples. For real-time operational deployment, fine-tuned encoders are therefore preferable despite lower overall F1.

2.5.3 STREET_INTERSECTION: The Critical Geotag Entity

`STREET_INTERSECTION` represents the most valuable geospatial entity type for downstream geocoding. The near-universal failure of fine-tuned models (excepting Qwen2.5) is the most critical limitation of this study. Future work should investigate: (1) targeted data augmentation using synthetic intersection descriptions generated from city street graph databases; (2) back-translation between Kazakh and Russian to increase example diversity; and (3) hybrid architectures combining an encoder for structured identifiers with an LLM for complex geographic spans.

Chapter 3

Conclusion

In the course of this diploma thesis, a multimodal system for forecasting cross-currency price relationships based on machine learning and deep learning methods was developed and investigated. The relevance of the research is determined by the high volatility of the modern foreign exchange market, the complexity of relationships between financial indicators, and the need to improve the accuracy of short-term exchange rate forecasting under unstable market conditions.

Within the theoretical part of the work, modern approaches to financial time series forecasting were considered, including statistical methods, machine learning algorithms, recurrent neural networks, and Transformer-based architectures. A comparative analysis of modern studies in the field of financial forecasting, multimodal systems, and methods for processing multidimensional time series was conducted. The literature review showed that deep learning models and Transformer-based approaches have demonstrated high effectiveness in recent years, as they are able to capture complex nonlinear dependencies and the influence of external factors on the dynamics of financial indicators.

In the practical part of the work, a forecasting system for cross-currency relationships was implemented for the EUR/KZT and USD/KZT currency pairs. Financial time series and additional market indicators were used to build the models, including the US Dollar Index (DXY), Brent oil prices, and the VIX market volatility index. The use of additional financial factors made it possible to form a multimodal dataset reflecting the influence of different segments of the global financial market on exchange rate dynamics.

During the development of the system, data collection, synchronization, and preprocessing were performed. To improve forecasting quality, feature generation methods were implemented, including lag features, rolling statistics, return indicators, and calendar features. For neural network models, data standardization was performed while preventing information leakage between the training and test samples.

Several forecasting models were implemented and compared within the framework of the research. CatBoost was used as a machine learning model for tabular financial data. Temporal Fusion Transformer was applied as a deep learning architecture designed for multivariate time series forecasting. TimeXer was implemented as a Transformer-based architecture focused on modeling relationships between the target time series and external financial indicators.

The experimental results demonstrated that all implemented models are capable of forecasting short-term exchange rate dynamics with relatively high accuracy. At the same time, the forecasting task remains extremely difficult due to the high stochasticity and instability of financial markets. Among the individual models, Temporal Fusion Transformer demonstrated the best overall balance between forecasting accuracy and robustness. TimeXer also showed strong performance due to its ability to account for exogenous variables and relationships between financial indicators.

An ensemble forecasting approach was additionally developed in order to combine the advantages of different models and improve forecasting stability. The ensemble approach demonstrated more robust behavior in several experimental scenarios and reduced the influence of large prediction errors produced by individual models.

The obtained results confirm the feasibility of applying machine learning and Transformer-based architectures to forecasting cross-currency price relationships. The developed system can be used in tasks related to financial analysis, risk assessment, and support of decision-making in foreign exchange markets.

Further research may be focused on increasing the volume of training data, integrating additional macroeconomic indicators, improving ensemble methods, and applying larger Transformer architectures. Another promising direction is the use of multimodal systems that combine financial time series with textual information, including financial news and market sentiment data.